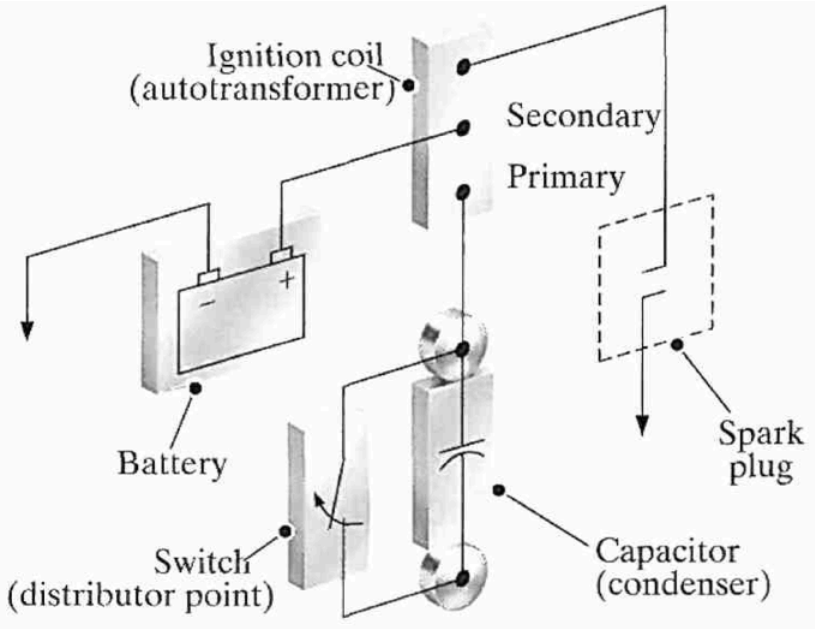
# Natural and Step Responses of RLC Circuits

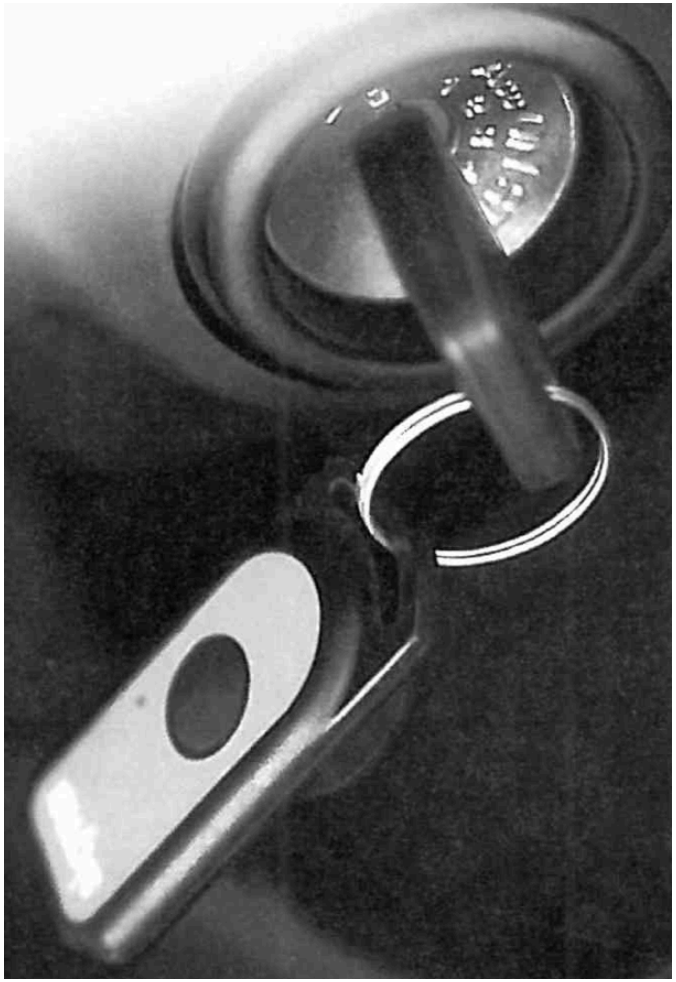
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Nov 2015

# Structure

- Introduction to the Natural Response of a Parallel *RLC* Circuit
- The Forms of the Natural Response of a Parallel *RLC* Circuit
- The Step Response of a Parallel *RLC* Circuit
- The Natural and Step Response of a Series *RLC* Circuit
- A Circuit with Two Integrating Amplifiers

# An Ignition Circuit

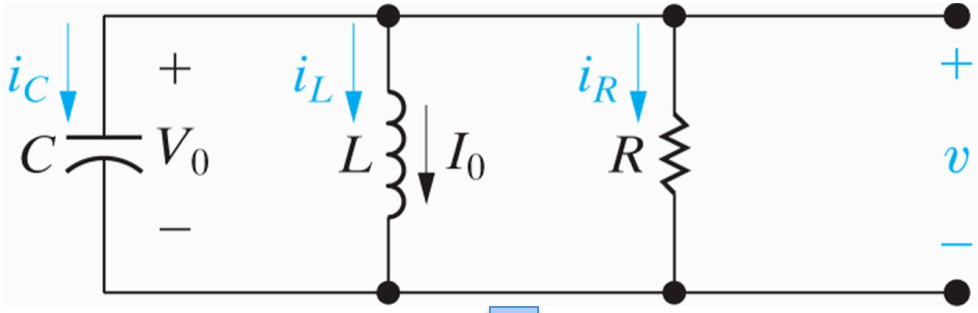


The rapidly changing current in the primary winding induces via magnetic coupling (mutual inductance) a very high voltage in the secondary winding.

# Introduction to the Natural Response of a Parallel *RLC* Circuit

The initial voltage on the capacitor, $V_0$ represents the initial energy stored in the capacitor. The initial current through the inductor, $I_0$ represents the initial energy stored in the inductor.




$$\frac{v}{R} + \frac{1}{L}\int_0^t v\,d\tau + I_0 + C\frac{dv}{dt} = 0$$

Second-ordered circuit

$$\frac{1}{R}\frac{dv}{dt} + \frac{v}{L} + C\frac{d^2v}{dt^2} = 0 \quad \Longleftarrow \quad \frac{d^2v}{dt^2} + \frac{1}{RC}\frac{dv}{dt} + \frac{v}{LC} = 0$$

# General Solution for the Second-Order Differential Equation

$$\frac{d^2v}{dt^2} + \frac{1}{RC}\frac{dv}{dt} + \frac{v}{LC} = 0$$

$$v = Ae^{st}$$

$$As^2e^{st} + \frac{As}{RC}e^{st} + \frac{Ae^{st}}{LC} = 0$$

$$s^2 + \frac{s}{RC} + \frac{1}{LC} = 0$$

Two roots:

$$s_1 = -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}}$$

$$s_2 = -\frac{1}{2RC} - \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}}$$

# *Some Useful Notations*

Denoting the two corresponding solutions $v_1$ and $v_2$, respectively, we can show that their sum also is a solution.

$$v = v_1 + v_2 = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$ (Proof from Eq. 8.10 to 8.12)

Only when $s_1 \neq s_2$

Denoting $\omega_0 = \dfrac{1}{\sqrt{LC}}$ $\alpha = \dfrac{1}{2RC}$ Unit: *radians per second* (rad/s)

We have $s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$ $s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$

| Parameter | Terminology | Value In Natural Response |
|---|---|---|
| $s_1, s_2$ | Characteristic roots | $s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$<br>$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$ |
| $\alpha$ | Neper frequency | $\alpha = \dfrac{1}{2RC}$ |
| $\omega_0$ | Resonant radian frequency | $\omega_0 = \dfrac{1}{\sqrt{LC}}$ |

**Over damped**: $\omega_0^2 < \alpha^2$ , both roots will be real and distinct

**Under damped**: $\omega_0^2 > \alpha^2$, two roots will be complex and conjungates

**Critical damped**: $\omega_0^2 = \alpha^2$, two roots will be real and equal

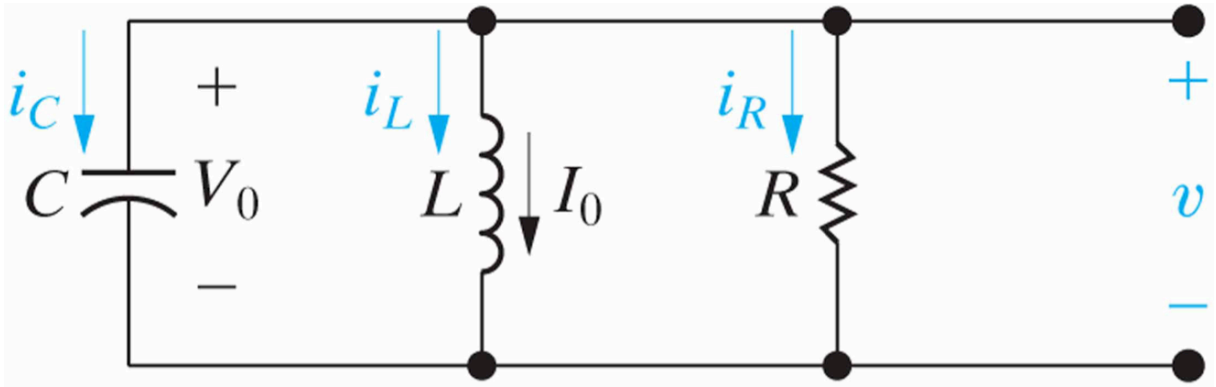
# Example #1

a) Find the roots of the characteristic equation that governs the transient behavior of the voltage if $R = 200\ \Omega$, $L = 50\ \text{mH}$, and $C = 0.2\ \mu F$.

b) Will the response be overdamped, underdamped, or critically damped?

c) Repeat (a) and (b) for $R = 312.5\ \Omega$.

d) What value of *R* causes the response to be critically damped?



# *Solution for Example #1*

a) For the given values of $R$, $L$, and $C$, we have

$$\alpha = \frac{1}{2RC} = \frac{10^6}{(400)(0.2)} = 1.25 \times 10^4 \text{ rad/s}$$

$$\omega_0^2 = \frac{1}{LC} = \frac{(10^3)(10^6)}{(50)(0.2)} = 10^8 \text{ rad}^2/\text{s}^2$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = 5000 \text{ rad/s}$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = 20000 \text{ rad/s}$$

b) In this case, $\alpha^2 > \omega_0{}^2$, thus the voltage response is *overdamped*.

c) Increase of $R$ will result in decrease of $\alpha$, thus the voltage response will be from *overdamped* to *underdamped*.

d) For critical damping, $\alpha^2 = \omega_0{}^2$, so

$$\left(\frac{1}{2RC}\right)^2 = \frac{1}{LC} = 10^8$$



$$\frac{1}{2RC} = 10^4$$



$$R = \frac{10^6}{(2 \times 10^4)(0.2)} = 250\ \Omega$$

# The Forms of the Natural Response of a parallel RLC Circuit

- Calculate the two roots $s_1$ and $s_2$, based on the given $R$, $L$ and $C$;
- Determine whether the response is over-, under-, or critically damped;
- Find the unknown coefficients, such as $A_1$ and $A_2$

# Overdamped Voltage Response

$$v = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$v(0^+) = A_1 + A_2 = V_0$$

$$\frac{dv(0^+)}{dt} = s_1 A_1 + s_2 A_2$$

$$\frac{dv(0^+)}{dt} = \frac{i_C(0^+)}{C} \qquad i_C(0^+) = \frac{-V_0}{R} - I_0$$

We summarize the process for finding the overdamped response, $v(t)$, as follows:

1. Find the roots of the characteristic equation, $s_1$ and $s_2$, using the values of $R$, $L$, and $C$.
2. Find $v(0^+)$ and $dv(0^+)/dt$ using circuit analysis.
3. Find the values of $A_1$ and $A_2$ by solving the following Equations simultaneously:

   $$v(0^+) = A_1 + A_2$$
   $$dv(0^+)/dt = i_C(0^+)/C = s_1A_1 + s_2A_2$$
4. Substitute the values for $s_1$, $s_2$, $A_1$, and $A_2$ into Eq. 8.18 to determine the expression for $v(t)$ for $t > 0$.

$$v = A_1e^{s_1t} + A_2e^{s_2t}$$

# Underdamped Voltage Response

$$s_1 = -\alpha + \sqrt{-(\omega_0^2 - \alpha^2)} = -\alpha + j\sqrt{\omega_0^2 - \alpha^2} = -\alpha + j\omega_d$$

$$s_2 = -\alpha - j\omega_d \rightarrow \omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

$$\begin{aligned} v(t) &= A_1 e^{(-\alpha + j\omega_d)t} + A_2 e^{-(\alpha + j\omega_d)t} \\ &= A_1 e^{-\alpha t} e^{j\omega_d t} + A_2 e^{-\alpha t} e^{-j\omega_d t} \rightarrow e^{\pm j\theta} = \cos\theta \pm j\sin\theta \\ &= e^{-\alpha t}(A_1 \cos\omega_d t + jA_1 \sin\omega_d t + A_2 \cos\omega_d t - jA_2 \sin\omega_d t) \\ &= e^{-\alpha t}[(A_1 + A_2)\cos\omega_d t + j(A_1 - A_2)\sin\omega_d t] \end{aligned}$$

$B_1$ Real! $B_2$

$$v(t) = B_1 e^{-\alpha t}\cos\omega_d t + B_2 e^{-\alpha t}\sin\omega_d t$$

$$v(0^+) = V_0 = B_1$$

$$\frac{dv(0^+)}{dt} = \frac{i_c(0^+)}{C} = -\alpha B_1 + \omega_d B_2$$

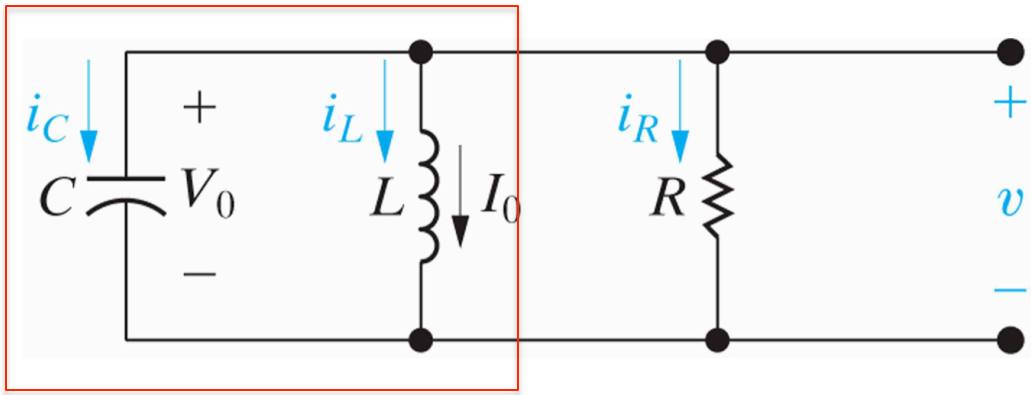
Because $\alpha$ determines how quickly the oscillations subside, it is also referred to as the damping factor (or damping coefficient).

The oscillatory behavior is possible because of the two types of energy-storage elements in the circuit: the *inductor* and the *capacitor*.

$R \rightarrow \infty, \quad \alpha \rightarrow 0$

The oscillation at $\omega_d = \omega_0$ is sustained!

No Dissipation on $R$.



# *Overdamped or Underdamped?*

- When specifying the desired response of a second order system, you may want to *reach the final value in the shortest time possible*, and you *may not be concerned with small oscillations about that final value*. If so, you would design the system components to achieve an **underdamped response**.
- On the other hand, you *may be concerned that the response not exceed its final value*, perhaps to *ensure that components are not damaged*. In such a case, you would design the system components to achieve an **overdamped response**, and you would *have to accept a relatively slow rise to the final value*.

# *Critically Damped Voltage Response*

$$s_1 = s_2 = -\alpha = -\frac{1}{2RC}$$

In the case of a repeated root, the solution involves a simple exponential term plus the product of a linear and an exponential term.

$$v(t) = D_1 t e^{-\alpha t} + D_2 e^{-\alpha t}$$

$$v(0^+) = V_0 = D_2$$

$$\frac{dv(0^+)}{dt} = \frac{i_C(0^+)}{C} = D_1 - \alpha D_2$$

You will rarely encounter critically damped systems in practice, largely because $\omega_0$ must equal $\alpha$ exactly. Both of these quantities depend on circuit parameters, and in a real circuit it is very difficult to choose component values that satisfy an exact equality relationship.

# Example #2

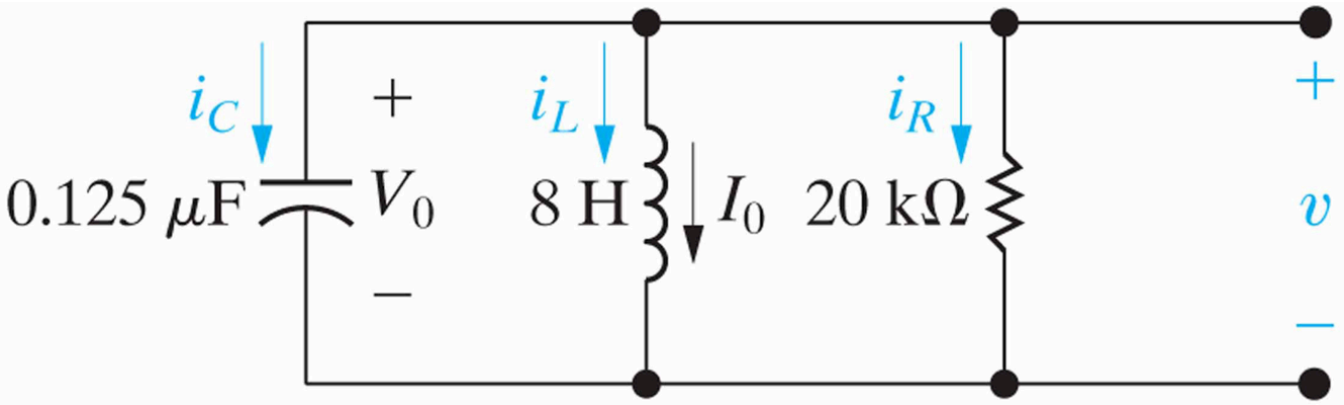
In the given circuit, $V_0 = 0$ and $I_0 = -12.25$ mA.

a) Calculate the roots of the characteristic equation.

b) Calculate $v$ and $dv/dt$ at $t = 0$.

c) Calculate the voltage response for $t \geq 0$.

d) Plot $v(t)$ versus $t$ for the time interval $0 \leq t \leq 11$ ms.



# Solution for Example #2

a)

$$\alpha = \frac{1}{2RC} = \frac{10^6}{2(20)10^3(0.125)} = 200 \text{ rad/s}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \sqrt{\frac{10^6}{(8)(0.125)}} = 10^3 \text{ rad/s}$$

$\omega_0^2 > \alpha^2$ Underdamped

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{10^6 - 4 \times 10^4} = 100\sqrt{96}$$

$$s_1 = -\alpha + j\omega_d = -200 + j979.80 \text{ rad/s}$$

$$s_2 = -\alpha - j\omega_d = -200 - j979.80 \text{ rad/s}$$

b)

$$v(0) = v(0^+) = V_0 = 0$$

$$i_C(0^+) = -(-12.25) = 12.25 \text{ mA}$$

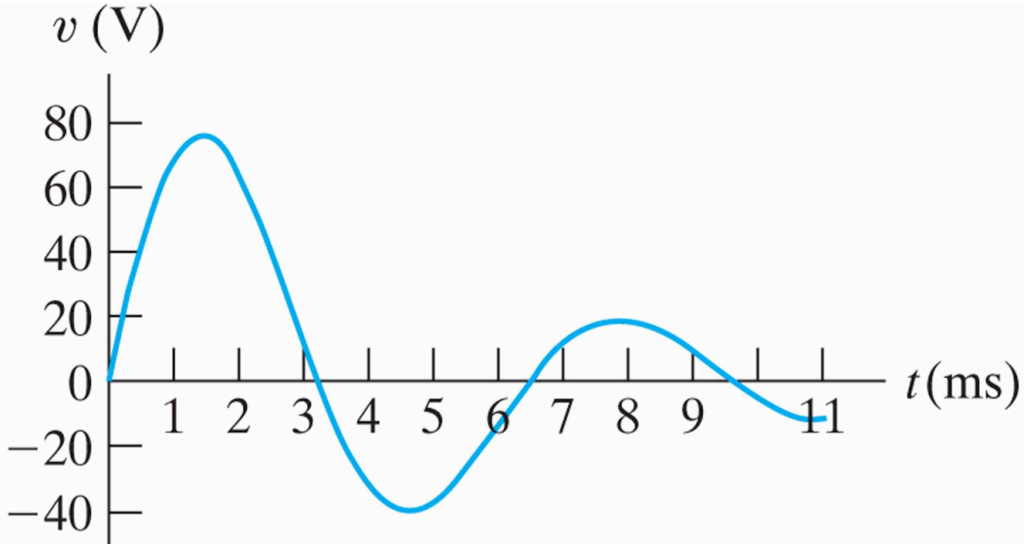
$$\frac{dv(0^+)}{dt} = \frac{(12.25)(10^{-3})}{(0.125)(10^{-6})} = 98{,}000 \text{ V/s}$$

$$v(0^+) = V_0 = B_1 = 0$$

$$\frac{dv(0^+)}{dt} = \frac{i_c(0^+)}{C} = -\alpha B_1 + \omega_d B_2 = 98{,}000 \text{ V/s}$$

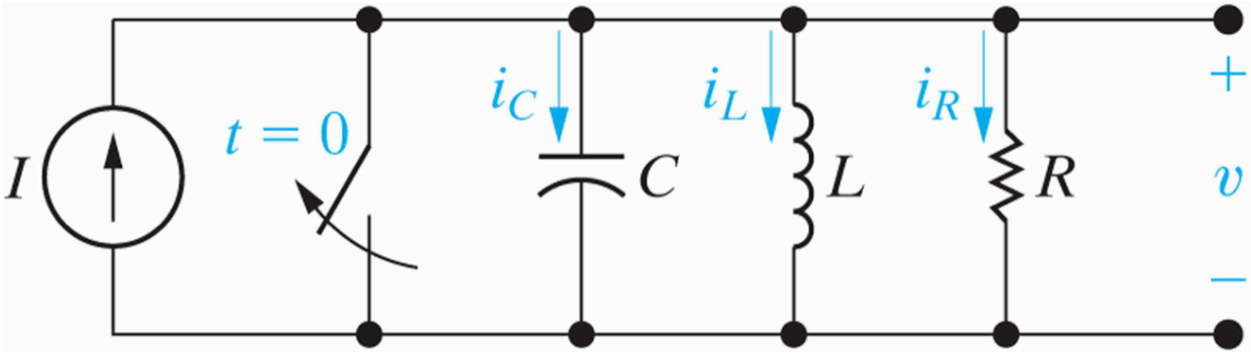
$$B_2 = \frac{98{,}000}{\omega_d} \approx 100 \text{ V}$$

$$v(t) = B_1 e^{-\alpha t} \cos \omega_d t + B_2 e^{-\alpha t} \sin \omega_d t$$
$$= 100 e^{-200t} \sin 979.80t \text{ V}, \quad t \geq 0$$



# Step Response of a Parallel *RLC* Circuit

$$i_L + i_R + i_C = I$$




$$i_L + \frac{v}{R} + C\frac{dv}{dt} = I$$

$$v = L\frac{di_L}{dt} \qquad \frac{dv}{dt} = L\frac{d^2 i_L}{dt^2}$$

$$i_L + \frac{L}{R}\frac{di_L}{dt} + LC\frac{d^2 i_L}{dt^2} = I \quad \Rightarrow \quad \frac{d^2 i_L}{dt^2} + \frac{1}{RC}\frac{di_L}{dt} + \frac{i_L}{LC} = \frac{I}{LC}$$

# The Indirect Approach

$$i_L + \frac{v}{R} + C\frac{dv}{dt} = I$$

$$\frac{1}{L}\int_0^t v\,d\tau + \frac{v}{R} + C\frac{dv}{dt} = I \quad \Rightarrow \quad \frac{d^2v}{dt^2} + \frac{1}{RC}\frac{dv}{dt} + \frac{v}{LC} = 0$$

$$v = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$
$$v = B_1 e^{-\alpha t}\cos\omega_d t + B_2 e^{-\alpha t}\sin\omega_d t$$
$$v = D_1 t e^{-\alpha t} + D_2 e^{-\alpha t}$$

$$i_L = I + A_1' e^{s_1 t} + A_2' e^{s_2 t}$$
$$i_L = I + B_1' e^{-\alpha t}\cos\omega_d t + B_2' e^{-\alpha t}\sin\omega_d t$$
$$i_L = I + D_1' t e^{-\alpha t} + D_2' e^{-\alpha t}$$

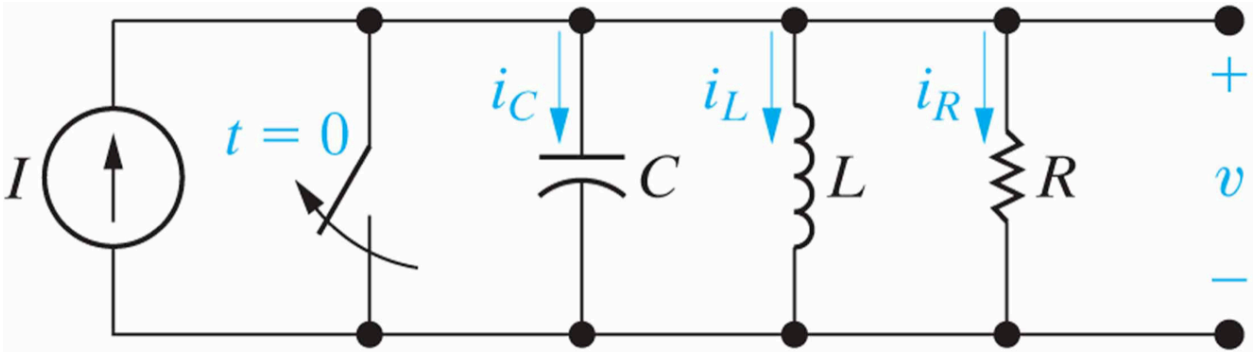
# The Direct Approach

$$i = I_f + \left\{ \begin{array}{c} \text{function of the same form} \\ \text{as the natural response} \end{array} \right\}$$

$= I$

$$v = V_f + \left\{ \begin{array}{c} \text{function of the same form} \\ \text{as the natural response} \end{array} \right\}$$

$= 0$



# Example #3

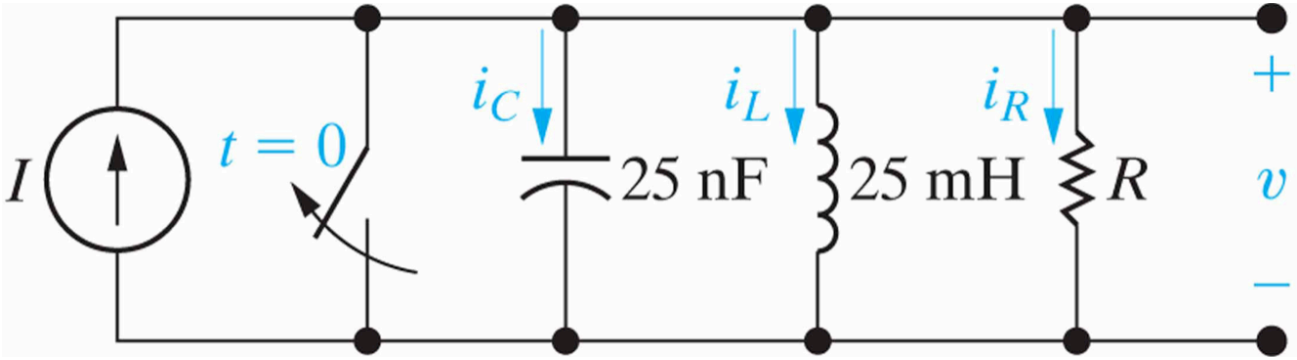
The initial energy stored in the given circuit is zero. At $t = 0$, a dc current source of $24\ \text{mA}$ is applied to the circuit. The value of the resistor is $400\ \Omega$.

a) What is the initial value of $i_L$?

b) What is the initial value of $di_L/dt$?

c) What are the roots of the characteristic equation?

d) What is the numerical expression for $i_L(t)$ when $t > 0$?



# *Solution for Example #3*

a) No energy is stored in the circuit prior to the application of the dc current source, so *the initial current in the inductor is zero*. The inductor prohibits an instantaneous change in inductor current; therefore $i_L(0) = 0$ immediately after the switch has been opened.

b) The *initial voltage on the capacitor is zero* before the switch has been opened; therefore it will be zero immediately after. Now, because $v = L di_L/dt$, we have

$$\frac{di_L}{dt}(0^+) = 0$$

c) From the circuit elements, we obtain

$$\omega_0^2 = \frac{1}{LC} = \frac{10^{12}}{(25)(25)} = 16 \times 10^8$$

$$\alpha = \frac{1}{2RC} = \frac{10^9}{(2)(400)(25)} = 5 \times 10^4 \text{ rad/s}$$

$\omega_0^2 < \alpha^2$ Overdamped

$$s_1 = -5 \times 10^4 + 3 \times 10^4 = -20{,}000 \text{ rad/s}$$
$$s_2 = -5 \times 10^4 - 3 \times 10^4 = -80{,}000 \text{ rad/s}$$

d) $i_L = I_f + A_1' e^{s_1 t} + A_2' e^{s_2 t}$

$$i_L(0) = I_f + A_1' + A_2' = 0$$
$$\frac{di_L}{dt}(0) = s_1 A_1' + s_2 A_2' = 0$$

$I_f = I = 24$ mA

$$A_1' = -32 \text{ mA}$$
$$A_2' = 8 \text{ mA}$$

$$i_L(t) = (24 - 32e^{-20{,}000t} + 8e^{-80{,}000t}) \text{ mA}, \quad t \geq 0$$

# The Natural Response of a Series RLC Circuit

$$Ri + L\frac{di}{dt} + \frac{1}{C}\int_0^t i\,d\tau + V_0 = 0$$

Differentiate

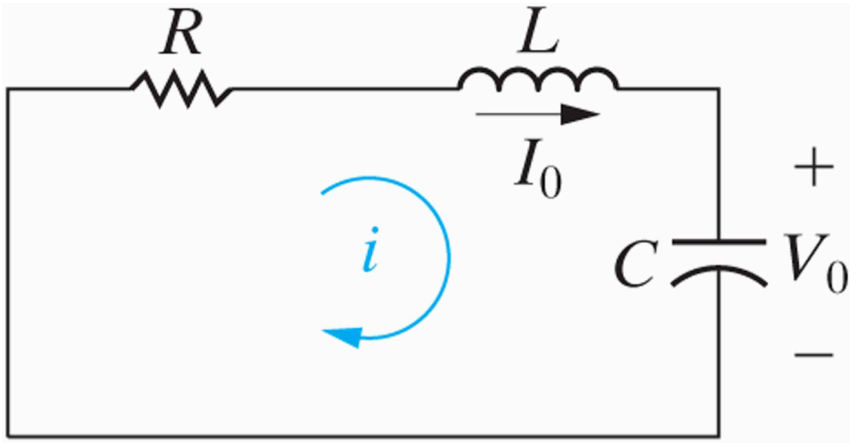
$$R\frac{di}{dt} + L\frac{d^2i}{dt^2} + \frac{i}{C} = 0$$

Rearrange

$$\frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{i}{LC} = 0$$

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

$$s_{1,2} = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$



$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s}$$

$$\alpha = \frac{R}{2L} \text{ rad/s}$$

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \text{ (overdamped)},$$

$$i(t) = B_1 e^{-\alpha t} \cos \omega_d t + B_2 e^{-\alpha t} \sin \omega_d t \text{ (underdamped)},$$

$$i(t) = D_1 t e^{-\alpha t} + D_2 e^{-\alpha t} \text{ (critically damped)}.$$

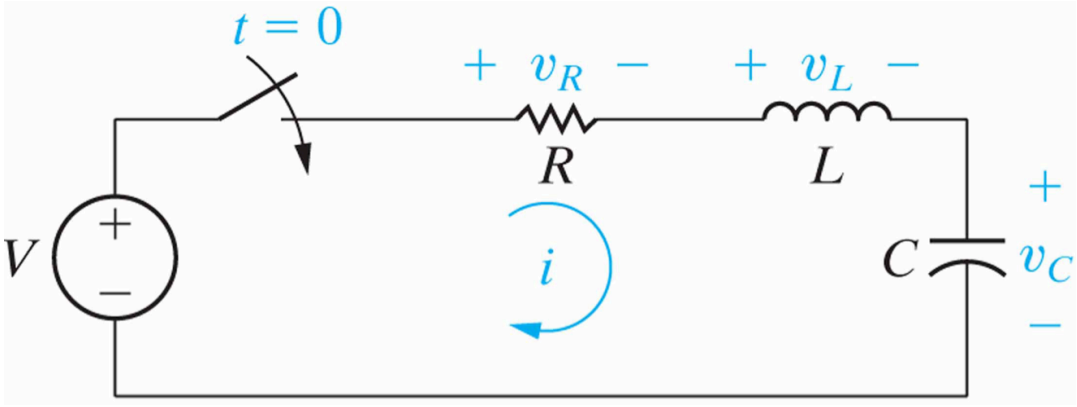
# The Step Response of a Series RLC Circuit

$$V = Ri + L\frac{di}{dt} + v_C$$

$$i = C\frac{dv_C}{dt}$$

$$\frac{di}{dt} = C\frac{d^2v_C}{dt^2}$$

$$\frac{d^2v_C}{dt^2} + \frac{R}{L}\frac{dv_C}{dt} + \frac{v_C}{LC} = \frac{V}{LC}$$




Compare Eq. 8.66 and Eq. 8.41:
the procedure for finding $v_C$ parallels that for finding $i_L$

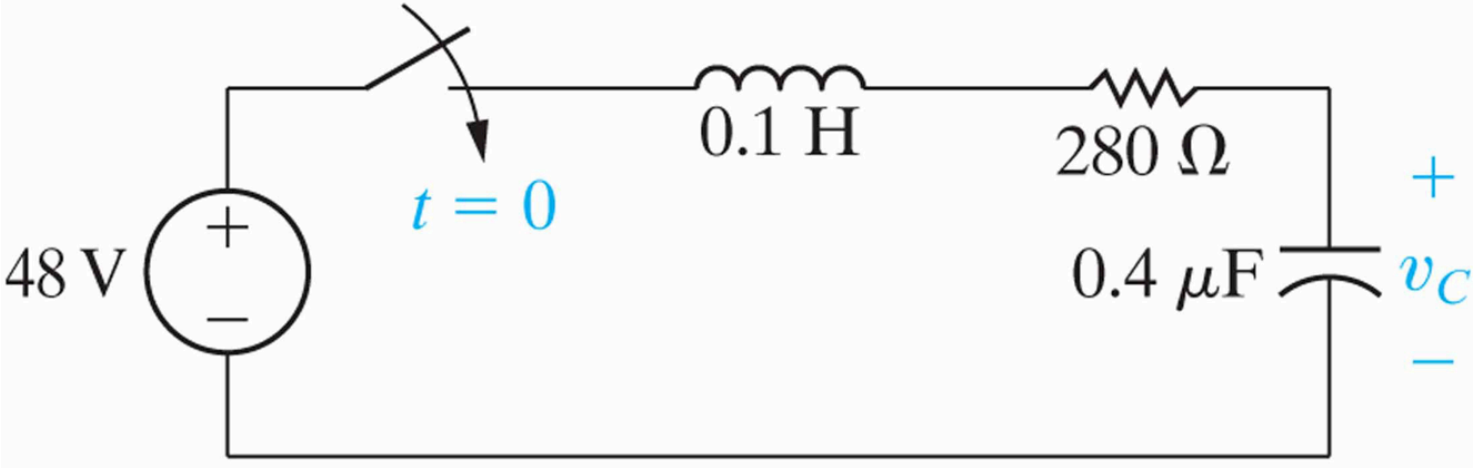
$$v_C = V_f + A'_1 e^{s_1 t} + A'_2 e^{s_2 t} \quad \text{(overdamped)},$$

$$v_C = V_f + B'_1 e^{-\alpha t}\cos \omega_d t + B'_2 e^{-\alpha t}\sin \omega_d t \quad \text{(underdamped)},$$

$$v_C = V_f + D'_1 t e^{-\alpha t} + D'_2 e^{-\alpha t} \quad \text{(critically damped)},$$

# Example #4

No energy is stored in the $100\ \text{mH}$ inductor or the $0.4$ $\mu\text{F}$ capacitor when the switch in the given circuit is closed. Find $v_C(t)$ for $t > 0$.



# Solution for Example #4

Calculate the two roots:

$$s_1 = -\frac{280}{0.2} + \sqrt{\left(\frac{280}{0.2}\right)^2 - \frac{10^6}{(0.1)(0.4)}} = (-1400 + j4800)\ \text{rad/s}$$

$$s_2 = (-1400 - j4800)\ \text{rad/s}$$

The roots are complex, so the voltage response is underdamped. Thus, we have

$$v_C(t) = 48 + B_1' e^{-1400t}\cos 4800t + B_2' e^{-1400t}\sin 4800t, \quad t \geq 0$$

No energy is stored in the circuit initially, so both $v_C(0)$ and $dv_C(0^+)/dt$ are zero, then

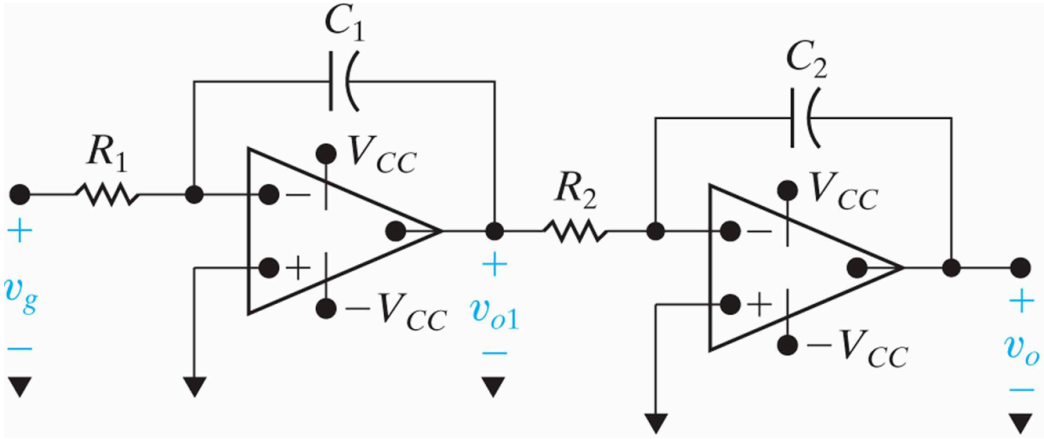
$$v_C(0) = 0 = 48 + B_1' \qquad \frac{dv_C(0^+)}{dt} = 0 = 4800B_2' - 1400B_1'$$

$$B_1' = -48\ \text{V} \quad B_2' = -14\ \text{V}$$

$$v_C(t) = (48 - 48e^{-1400t}\cos 4800t - 14e^{-1400t}\sin 4800t)\ \text{V}, \quad t \geq 0$$

# A Circuit with Two Integrating Amplifier

Two integrating amplifiers connected in cascade: the output signal of the first amplifier is the input signal for the second amplifier.




For the first integrator:

$$\frac{0 - v_g}{R_1} + C_1\frac{d}{dt}(0 - v_{o1}) = 0 \quad \Rightarrow \quad \frac{dv_{o1}}{dt} = -\frac{1}{R_1C_1}v_g$$

For the second integrator:

$$\frac{dv_o}{dt} = -\frac{1}{R_2C_2}v_{o1} \quad \Rightarrow \quad \frac{d^2v_o}{dt^2} = -\frac{1}{R_2C_2}\frac{dv_{o1}}{dt}$$

$$\Rightarrow \quad \frac{d^2v_o}{dt^2} = \frac{1}{R_1C_1}\frac{1}{R_2C_2}v_g$$

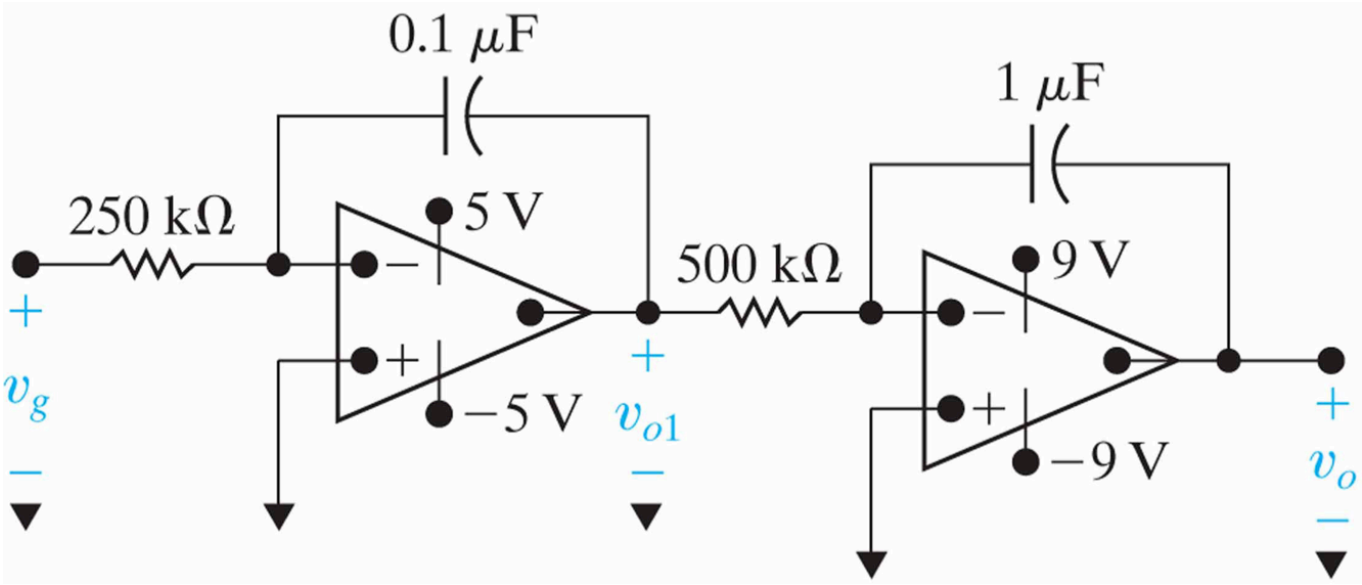
Second-order

# Example #5

No energy is stored in the given circuit when the input voltage $v_g$ jumps instantaneously from 0 to 25 mV.

a) Derive the expression for $v_o(t)$ for $0 < t < t_{\text{sat}}$.

b) How long is it before the circuit saturates?



# Solution for Example #5

a)

$$\frac{1}{R_1C_1} = \frac{1000}{(250)(0.1)} = 40$$

$$\frac{1}{R_2C_2} = \frac{1000}{(500)(1)} = 2$$

$v_g = 25$ mV

$$\frac{d^2v_o}{dt^2} = (40)(2)(25 \times 10^{-3}) = 2$$

$$g(t) = \frac{dv_o}{dt}$$

Integrate

$$dg(t) = 2dt$$

$$\int_{g(0)}^{g(t)} dy = 2\int_0^t dx$$

$$g(t) - g(0) = 2t$$

$$g(0) = \frac{dv_o(0)}{dt} = 0$$

$$\frac{dv_o}{dt} = 2t$$

$$v_o(0) = 0$$

$$v_o = t^2, \quad 0 \le t \le t_{sat}$$

b) When t = 3 s, the second amplifier saturates, at that time we get:

$$\frac{dv_{o1}}{dt} = -40(25) \times 10^{-3} = -1$$

$$v_{o1} = -t = -3 \text{ V} > -5 \text{ V}$$

The first amplifier doesn't saturates!

# Summary

- characteristic equation
- Second-order differential equation
- Overdamped, underdamped, and critical damped
- Natural Step responses